

Sydney Technical High School



Mathematics

H.S.C. ASSESSMENT TASK 3

JUNE 2012

General Instructions

- Working Time – 70 minutes.
- Approved calculators may be used.
- A table of Standard Integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a new side of the answer booklet.
- Marks shown are a guide and may need to be adjusted.
- Full marks may not be awarded for careless work or illegible writing.
- For Questions 1-5, write the letter for the correct answer on the first page of your answer booklet. Be very clear.

NAME _____

TEACHER _____

Question 1

The sine curve with period 4π units and amplitude 2 units has equation:

- A. $y = 2 \sin \frac{x}{2}$ B. $y = 2 \sin \frac{x}{4}$ C. $y = 4 \sin 2x$ D. $y = 4 \sin \frac{x}{2}$ E. $y = 2 \sin 4x$

Question 2

The derivative of $\sin^2 x$ is :

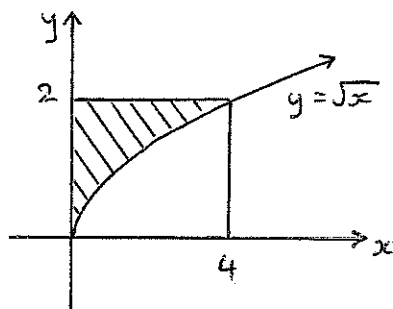
- A. $\cos^2 x$ B. $2 \sin x$ C. $2 \cos x$ D. $2 \cos x \sin x$ E. none of these.

Question 3

The primitive of $\cos^2 x$ is :

- A. $\sin^2 x$ B. $\frac{\cos^3 x}{3}$ C. $\frac{\sin^3 x}{3}$ D. $\frac{\cos^3 x}{3 \sin x}$ E. none of these.

Question 4

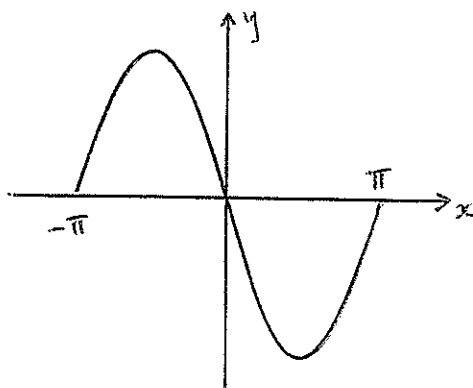


The shaded area can be found using :

- A. $\int_0^2 \sqrt{x} \, dx$ B. $\int_0^2 y \, dy$ C. $\int_0^2 y^2 \, dy$
D. $\int_0^4 y^2 \, dy$ E. $\int_0^4 \sqrt{x} \, dx$

Question 5

Which of the following is NOT a possible function for the curve shown :



- A. $y = -\sin x$ B. $y = \sin(x + \pi)$
C. $y = \sin(x - \pi)$ D. $y = \cos(x - \frac{\pi}{2})$
E. $y = \cos(x + \frac{\pi}{2})$

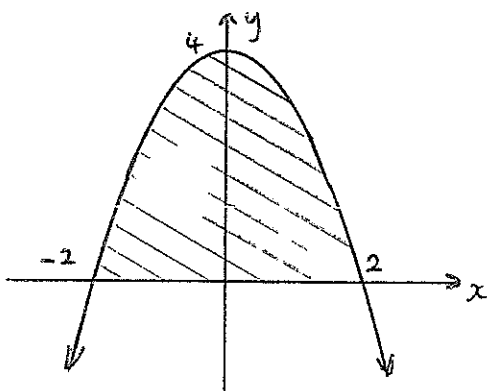
Question 6 (12 marks) Start on a new page.

Marks

- a) Convert $\frac{\pi}{10}$ radians to degrees. 1
- b) Give the exact value of $\operatorname{cosec} \frac{\pi}{4}$ 1
- c) Solve $\tan^2 x - \tan x = 0$ for $0 \leq x \leq 2\pi$ 3
- d) Find the gradient of the tangent to the curve $y = 3 \sin 2x$ at the point where $x = \frac{\pi}{12}$ 2
- e) Evaluate $\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \sec^2 2x \, dx$. Leave your answer in exact form. 2
- f) Find the total area between the curve $y = \sin x$ and the x axis for $-\frac{3\pi}{2} \leq x \leq \frac{3\pi}{2}$ 2
- g) Evaluate $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \tan x \, dx$ 1

Question 7 (14 marks) Start on a new page.

- a) i) Find $\frac{d}{dx}(\tan^2 x)$ 1
- ii) Hence find $\int \tan x \sec^2 x \, dx$ 1
- b) Find $\frac{d}{dx}(\cos^3 5x)$ 2
- c) The graph of $y = 4 - x^2$ is shown :

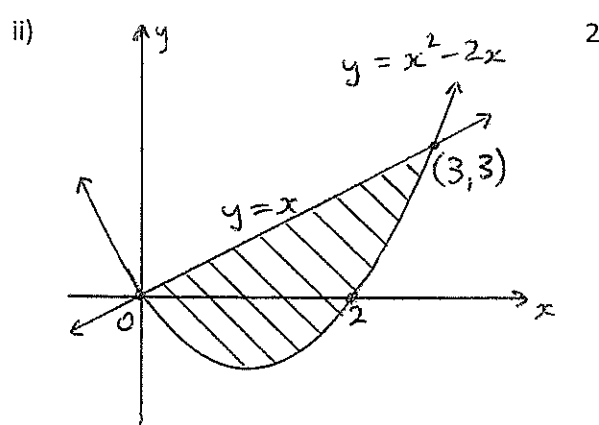
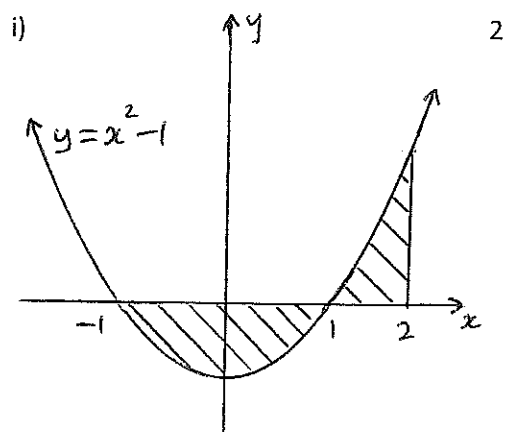


- i) Use the Trapezoidal Rule and 5 function values to approximate the shaded area above. 2
- ii) Find the exact value of the shaded area. 3
- iii) The shaded area is rotated about the y -axis. Find the generated volume in exact form. 3
- d) Find an angle, x radians, such that the gradient on the curve $y = \tan x$ has value 2. 2

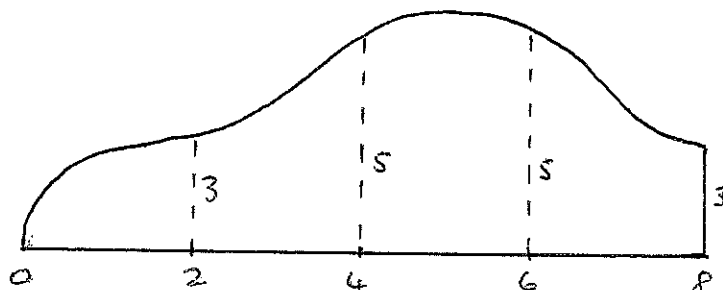
Question 8 (12 marks) Start on a new page.

a) Write an integral expression that represents the total shaded area of each situation below :

DO NOT EVALUATE THE INTEGRALS.



b) The cross-sectional area of a rock wall is shown. Horizontal lengths and their corresponding vertical heights are indicated, in metres.



Find the approximate area above using Simpson's Rule and 5 function values. 2

c) i) Sketch the curve $y = 2 \cos 4x$ for $0 \leq x \leq \frac{\pi}{4}$. Use a ruler and clearly label x, y intercepts. 2

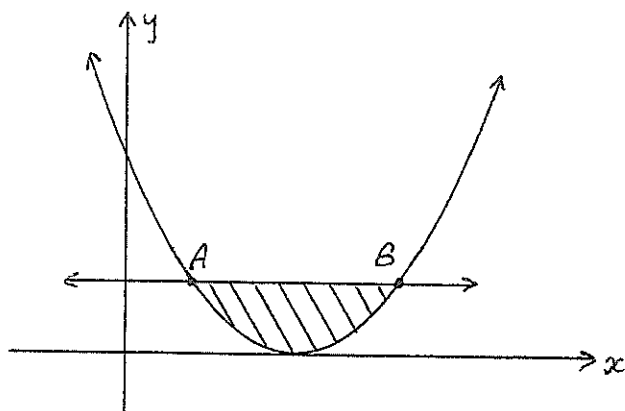
ii) Evaluate $\int_0^{\frac{\pi}{8}} 2 \cos 4x \, dx$ 2

iii) On the same axes as i), draw the line $y = 2x$ 1

iv) Use your graphs above to estimate the solution to $x - \cos 4x = 0$, in terms of π . 1

Question 9 (13 marks) Start on a new page.

- a) The area between the graphs of $y = (x - 2)^2$ and $y = 1$ is shown.



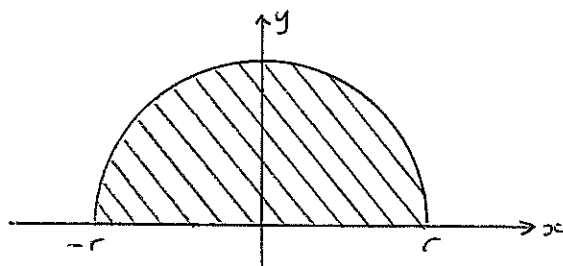
- i) Find x values for A and B .

2

- ii) Find the shaded area.

3

- b) The area between the semi-circle $y = \sqrt{r^2 - x^2}$ and the x -axis is shown.



- i) Evaluate $\int_{-r}^r \sqrt{r^2 - x^2} \, dx$

1

- ii) The shaded area is rotated about the x -axis. Use calculus to find the exact volume thus generated.

3

- c) i) Show that $\frac{d}{dx}(\sin^3 x) = 3 \cos x - 3 \cos^3 x$

2

- ii) Hence find $\int \cos^3 x \, dx$

2

END OF TEST

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Solutions.

① A ② D ③ E ④ C ⑤ D

⑥ a) 18° b) $\frac{1}{\sin \frac{\pi}{4}} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$ c) $\tan x (\tan x - 1) = 0$
 $\tan x = 0$ or 1
 $\therefore x = 0, \pi, 2\pi, \frac{\pi}{4}, \frac{5\pi}{4}$

d) $y' = 3 \cos 2x \times 2$ At $x = \frac{\pi}{12}$, $m_T = 6 \cos \frac{\pi}{6}$
 $= 6 \cos 2x = 6 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$

e) $\left[\frac{\tan 2x}{2} \right]_{\frac{\pi}{8}}^{\frac{\pi}{6}} = \frac{\tan \frac{\pi}{3}}{2} - \frac{\tan \frac{\pi}{4}}{2}$ f) 
 $= \frac{\frac{\sqrt{3}}{2}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$
 $\text{Area} = 3 \times \int_0^{\pi} \sin x \, dx$
 $= -3 [\cos x]_0^{\pi}$
 $= -3(-1-1) = 6u^2$

g) 0

⑦ a) i) $2 \tan x \sec^2 x$ ii) $\frac{1}{2} \tan^2 x + c$

b) $3(\cos 5x)^2 \times -\sin 5x \times 5 = -15 \cos^2 5x \sin 5x$

c) i) $\text{Area} \div 2 \times \left[\frac{1-0}{2}(4+3) + \frac{2-1}{2}(3+0) \right]$ ii) $\text{Area} = 2 \times \int_0^2 (4-x^2) dx$
 $= 2 \times \frac{1}{2}(10) = 10u^2$
 $= 2 \left[4x - \frac{x^3}{3} \right]_0^2$
 $= 2(8 - \frac{8}{3}) = 2 \times 5\frac{1}{3} = 10\frac{2}{3} u^2$

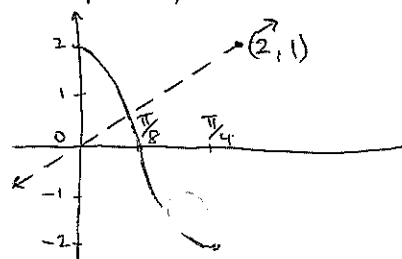
iii) $x^2 = 4-y$
 $\therefore \text{Vol} = \pi \int_0^4 (4-y) dy$
 $= \pi \left[4y - \frac{y^2}{2} \right]_0^4$
 $= \pi(16-8) = 8\pi u^3$

d) $\frac{d}{dx} (\tan x) = 2$
 $\therefore \sec^2 x = 2$
 $\frac{1}{\cos^2 x} = 2$
 $\cos^2 x = \frac{1}{2}$
 $\cos x = \pm \frac{1}{\sqrt{2}}$
 $\therefore x = \frac{\pi}{4} \text{ (say)}$

⑧ a) i) $A = \left| \int_{-1}^1 (x^2-1) dx + \int_1^2 (x^2-1) dx \right|$
 $\equiv 2 \left| \int_0^1 (x^2-1) dx \right| + \int_1^2 (x^2-1) dx$
 ii) $A = \left| \int_0^3 (x^2-2x-x) dx \right|$
 $= \left| \int_0^3 (x^2-3x) dx \right|$
 $\equiv \int_0^3 (3x-x^2) dx$

b) $\text{Area} \div \frac{2}{3} (0+4 \times 3 + 2 \times 5 + 4 \times 5 + 3)$
 $= \frac{2}{3} (45) = 30m^2$

c) i) amp. = 2, period = $\frac{2\pi}{4} = \frac{\pi}{2}$



ii) $\int_0^{\frac{\pi}{8}} 2 \cos 4x \, dx$
 $= \left[\frac{\sin 4x}{2} \right]_0^{\frac{\pi}{8}}$
 $= \frac{1}{2} - \frac{0}{2} = \frac{1}{2}$

iii) on graph
 iv) same as solving
 $2x = 2 \cos 4x$
 i.e. approx.
 $x = \frac{\pi}{9}$
 (say)

⑨ a) i) $(x-2)^2 = 1$
 $x^2 - 4x + 4 = 1$
 $x^2 - 4x + 3 = 0$
 $(x-1)(x-3) = 0$
 $\therefore x = 1 \text{ or } 3$

ii) $\text{Area} = \text{rectangle} - \text{area under arc}$
 $= 2 \times 1 - \int_1^3 (x-2)^2 dx$
 $= 2 - \left[\frac{(x-2)^3}{3} \right]_1^3$
 $= 2 - \left(\frac{1}{3} - -\frac{1}{3} \right)$
 $= 2 - \frac{2}{3} = 1\frac{1}{3} u^2$

$$b) \quad i) A = \frac{1}{2} \text{ circle} \\ = \frac{\pi r^2}{2}$$

$$\begin{aligned} ii) Vol &= 2\pi \int_0^r (\sqrt{r^2 - x^2})^2 dx \\ &= 2\pi \int_0^r (r^2 - x^2) dx \\ &= 2\pi \left[r^2 x - \frac{x^3}{3} \right]_0^r \\ &= 2\pi \left[(r^3 - \frac{r^3}{3}) - (0 - 0) \right] \\ &= 2\pi \times \frac{2r^3}{3} \\ &= \frac{4\pi r^3}{3} \quad u^3 \end{aligned}$$

$$\begin{aligned} c) \quad i) \frac{d}{dx} [(\sin x)^3] &= 3(\sin x)^2 \times \cos x \\ &= 3 \sin^2 x \cos x \\ &= 3(1 - \cos^2 x) \cos x \\ &= 3 \cos x - 3 \cos^3 x \quad \text{as reqd.} \end{aligned}$$

$$ii) 3 \cos^3 x = 3 \cos x - \frac{d}{dx} (\sin^3 x)$$

$$\therefore \cos^3 x = \cos x - \frac{1}{3} \frac{d}{dx} (\sin^3 x)$$

$$\begin{aligned} \therefore \int \cos^3 x \, dx &= \int \cos x \, dx - \frac{1}{3} \int \frac{d}{dx} (\sin^3 x) \, dx \\ &= \sin x - \frac{1}{3} \sin^3 x + C \end{aligned}$$